

The Theory of Contraction of Proteins under their Excitation

A. S. DAVYDOV

*The Institute of Theoretical Physics,
Academy of Sciences of the Ukrainian S.S.R.,
Kiev, U.S.S.R.*

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This paper develops the quantum theory of contraction of α -helical proteins under the excitation of their peptide groups, which form three parallel chains of hydrogen bonds along the molecule. In the region embraced by an excitation, the pitch of the helix decreases, and the "contracted" region of the helix moves along the molecule at a rate proportional to the energy of resonance interaction between peptide groups. The movement of the excitation region from the head of the myosin molecule to its tail results in movement of the myosin filaments along the actin ones in the myofibril of the muscle, and thus leads to a decrease in length of the muscle fibre.

1. Introduction

As yet, there is no complete understanding of the mechanism for muscle contraction in vertebrates or invertebrates. Apparently it is an established fact that the energy required for the contraction of muscle fibres is released when adenosine triphosphate (ATP) transforms into adenosine diphosphate (ADP) and phosphate. Such a transformation occurs in the presence of Ca^{2+} ions, which trigger an enzyme action of the protein molecules of the muscles. The mechanism converting this chemical energy into a mechanical motion is still not completely understood. In the present paper we propose a new mechanism responsible for the contraction of protein molecules when they are excited by the energy released in the hydrolysis of ATP.

2. Current Concepts of the Mechanism for Muscle Contraction

As is well known, long polymer protein molecules (polypeptides) are the chief contractile elements of muscles and other biological objects that con-

vert chemical energy into mechanical motion. The striated muscles of vertebrate animals and insects have received a very detailed study (see, for example, the review of Volkenstein (1970)). These muscles are bundles of fibres each of which is made up of several thousands of densely packed filaments called myofibrils. The myofibril is the functional unit of the muscle contractile system.

Each myofibril is divided into separate strictly repeating segments, or sarcomeres, built up of an interpenetrating system of thin (actin) and thick (myosin) protein filaments. The thick filaments are situated in the centre part of the sarcomeres and consist of several hundred myosin molecules. These molecules are built up of two identical polypeptide chains coiled up in the form of a helix. In turn, each of these chains has the form of an α -helix in the long (tail) portion of the molecule. In the head portion the molecule has two projections ("heads"). Separate myosin molecules in thick filaments are oriented in opposite directions from the centre.

The thin filaments of the sarcomere start from the Z-plates (which separate the sarcomeres from each other) and run to the centre of the sarcomere, partially penetrating into the spaces between thick filaments. They are made up of double helices of globular G-actin molecules. Each filament contains about 400 molecules.

Many hypotheses have been proposed to account for the machinery of contraction. Some of these hypotheses were based on the properties of polyelectrolytes which lead to the folding and unfolding of protein chains when the charges of their radicals are changed. These hypotheses differed from each other in the assumptions concerning the character of the distribution and the change in electric charges along the protein molecules (Meyer, 1929; Kuhn, Toth & Ebner, 1962; Wohlsch, 1940; Botts, Morales & Cellular, 1951; Risemann & Kirkwood, 1948). No direct evidence is available for the rearrangement of molecules during the contraction of muscle fibres.

Hypotheses have often been advanced (Astbury, 1950; Huxley, A., 1957; Ooswa *et al.*, 1961) to account for the contraction and relaxation of muscles by postulating the structural transformation of myosin and actin molecules.

A number of authors (Pryor, 1950; Kuhn, 1960; Flory, 1956; Hill, 1968) have associated muscle contraction with phase transitions of the protein molecules. But the bulk of X-ray evidence on living contracting muscle does not support the hypothesis that there is any large scale rearrangement in the filaments of muscle (see review paper by J. Hanson (1968)).

Using electron-microscope data, Huxley H. & Hanson (1954) and Huxley & Niedergerke (1954) established that the contraction of myofibrils was connected with the fact that thin actin filaments move along the channels between the thick myosin filaments without overall changes in length. These

data served as a basis in constructing the presently widely accepted "sliding model" for the mechanism of muscle contraction and relaxation.

A phenomenological theory of the sliding model has been developed by Davies (1963). Davies' theory runs into a number of difficulties. It overestimates the role of electrostatic repulsion, regarding it as causing the myosin molecule's head to change from an α -helical to a stretched, disordered configuration.

The distance between the G-actin monomers in the actin filament is about 55 Å. Consequently, every miniature contraction cycle should, according to Davies' theory, move the actin filament ~ 55 Å farther, so that the molecule's head should shorten and stretch by about 70 Å during each cycle.

If the negative electric charge at the base of the myosin molecule's head so strongly repels the ATP^- complex, attached to the head, that the latter stretches, then the same charge will not allow the sarcoplasmic ATP to approach the ADP (which formed at the head after the ATP broke down) to replace it or charge it to ATP.

Davies' theory does not make it clear why the myosin molecule has, besides its head, a very long (~ 1300 Å) α -helical tail that takes no active part in the contraction. It is unlikely that nature is so wasteful.

3. Collective Excitations of Peptide Groups in α -Helical Protein Molecules

The contractile proteins, like all other proteins, are synthesized by a linear polymerization of various amino acids. Polymerization results in a periodically repeating sequence of peptide groups



The formation of hydrogen bonds between the oxygen and hydrogen atoms of the peptide groups of the first and fourth, second and fifth, etc., amino acid residues stimulates the polypeptide chains of amino acid residues to fold up into α -helices. Thus there forms an α -helix 10.1 Å in diameter, containing three chains of hydrogen bonds whose directions almost coincide with that of the helix axis.

As the peptide groups are regularly positioned, collective (exciton) excitations forming broad bands correspond to electric dipole excitations. Moffitt (1956), using the theory of molecular crystal spectra developed by

Davydov (1951, 1971), developed the theory of collective excitations of a type which correspond in energy range (5–7 eV) to the electronic excitations of peptide groups. In calculations it was assumed that the peptide groups were fixed in the positions which they occupy in a non-excited molecule.

As this author has shown (1969), if the deformability of a crystal lattice is taken into account one is led to the fact that an exciton excitation in a one-dimensional crystal is accompanied by a local crystal deformation. We shall consider this effect for the case where α -helical molecules are subject to excitations of less than the 0.48 eV of energy released during ATP hydrolysis. This energy range corresponds to the energy of 0.23 eV of the valence vibration C=O in the peptide group. The electric dipole moment of this vibration is directed along the hydrogen bond. The dipole–dipole interaction between adjacent peptide groups placed along the chain of hydrogen bonds is an order of magnitude higher than the similar interaction between the nearest groups of adjacent chains, so that the excitations of the peptide groups of the three hydrogen-bond chains may be treated independently. For the sake of simplicity we shall consider only one of them in what follows.

Let $\mathbf{r}_n$ be a radius-vector determining the position of the n th peptide group (see Fig. 1). The number n takes the values 1, 2, ..., N ($N \sim 200$ –300). Then

$$R_n = |\mathbf{r}_{n+1} - \mathbf{r}_n| \quad (2)$$

is the distance between two adjacent peptide groups. Suppose that the potential interaction energy of these groups when in the ground state is

$$U_0 = \sum_n u_0(R_n), \quad (3)$$

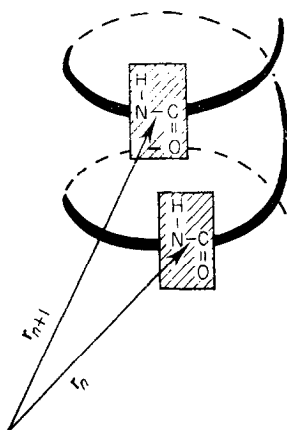


FIG. 1.

where

$$u_0(R) = C/R^s - A/R^3, \quad (4)$$

The quantities C , A and s are the parameters of the theory. They depend on the properties of peptide groups themselves and the environment in which they happen to be (cytoplasm, radicals, etc.). To the minimum of potential energy, equation (3), there correspond the equilibrium distances

$$R_0 = (sC/3A)^\sigma, \quad \sigma = (s-3)^{-1}. \quad (5)$$

The value of R_0 is of the order of $5.4 \text{ \AA}$.

Suppose that the peptide group in a protein molecule can go into an excited state with excitation energy ε , and that an electric dipole transition moment directed along the molecule corresponds to this excitation.

Let us introduce the operators of creation B_n^+ and annihilation B_n of an excitation on group n . Then for fixed values of R_n the operator for the excitation energy of the whole chain will have the form

$$H = \sum_n \{[\varepsilon + D(R_n)]B_n^+ B_n + M(R_n)[B_n B_{n+1}^+ + B_n^+ B_{n+1}]\}, \quad (6)$$

where

$$D(R) = -\alpha/R^3 \quad (7)$$

is the change in static interaction energy of two adjacent groups when one of them goes into an excited state:

$$M(R) = -\gamma/R^3 \quad (8)$$

is the energy of resonance dipole-dipole interactions between adjacent groups. $\gamma = 2\mathbf{d}; \mathbf{d}$ is the electric moment of the dipole excitation.

To simplify our calculation we take periodic boundary conditions. The $R_N = |\mathbf{r}_1 - \mathbf{r}_N|$ and the sums over n in all formulae are extended over the values from 1 to N .

If the ground state of the chain is $|0\rangle$, then at fixed R_n the excited states are characterized by the functions

$$\Psi = \sum_n a_n B_n^+ |0\rangle \quad (9)$$

which are defined by the sets of complex numbers a_n satisfying the condition

$$\sum_n a_n^* a_n = 1. \quad (10)$$

To every set of numbers a_n and R_n there corresponds the excitation energy

$$E(\{a_n\}, \{R_n\}) = \sum_n \{[\varepsilon + D(R_n)]a_n^* a_n + M(R_n)[a_n a_{n+1}^* + a_n^* a_{n+1}]\}. \quad (11)$$

For a fixed set of numbers $\{a_n\}$, the expression (11) is dependent on the distance R_n and is the potential energy, additional to that in equation (3), for peptide groups.

Hence the total potential energy takes the form

$$U(\{a_n\}, \{R_n\}) = \varepsilon + \sum_n \{u_0(R_n) + D(R_n)a_n^*a_n + M(R_n)[a_na_{n+1}^* + a_n^*a_{n+1}]\}. \quad (12)$$

Our task is to find sets a_n and R_n corresponding to an absolute minimum of the energy given in equation (12), at a fixed number of simultaneous elementary excitations—that is, when the condition (10) is satisfied.

We obtain the minimization of the energy, (formula (12)), in two stages. We first find the values $R_n = \rho_n$ that correspond to the minimum of formula for a fixed set of values of the a_n . These values of R_n determine new equilibrium distances between the peptide groups in an excited chain:

$$\rho_n = R_0 \{1 + \kappa a_n^*a_n + \eta[a_na_{n+1}^* + a_n^*a_{n+1}]\}^{-\sigma}, \quad (13)$$

where

$$\kappa = \frac{\alpha}{A}, \quad \eta = \frac{\gamma}{A} > 0 \quad (\eta < \kappa < 1). \quad (14)$$

To find the coefficients a_n which minimize formula (12), we insert the values (13) and keep the terms of order not higher than κ, η, κ^2 . As a result we obtain the functional

$$U(\{a_n\}) = Nw_0 + \varepsilon - (3\sigma + 1)\kappa\gamma w_0 - \Lambda \sum_n (a_na_{n+1}^* + a_n^*a_{n+1}) - G \sum_n (a_n^*a_n)^2, \quad (15)$$

where

$$w_0 = -u_0(R_0) = \frac{A}{\sigma SR_0^3}, \quad G = \frac{3}{2}(\sigma + 1)\sigma\kappa^2 w_0, \quad \Lambda = \frac{(3\sigma + 1)\gamma}{\sigma SR_0^3} \quad (16)$$

are parameters independent of n . With our choice of boundary conditions the potential energy (equation (12)) is therefore invariant under the substitution of $n+m$ for n , where m is an integer. Consequently, the energy states of the chain may be classified by means of the wave numbers

$$k = \frac{2\pi}{N\rho_0} j, \quad (j = 0, \pm 1, \pm \dots N/2), \quad (17)$$

where ρ_0 is the average distance between peptide groups in the region of excitation. Considering only small values of k (long wave approximation), one can go over in equation (12) to a continuous model of the chain.

4. Collective Excitations in Long Wave Approximation

To go over to a continuous model of the chain in the case of long wave excitation we perform the substitution

$$1 = \sum_n a_n^*a_n \Rightarrow \int a^*(x)a(x) dx, \quad (18)$$

$$\varepsilon - \Lambda \sum_n (a_n a_{n+1}^* + a_n^* a_{n+1}) \Rightarrow (\varepsilon - 2\Lambda) + \int a^*(x) \left(-\frac{\hbar^2}{2m^* \rho_0^2} \frac{d^2}{dx^2} \right) a(x) dx, \quad (19)$$

where

$$m^* = \hbar^2 / (2\Lambda \rho_0^2) \quad (20)$$

is the effective mass of a free exciton. Then equation (15) reduces to the form

$$U(k, \{a(x)\}) = (\varepsilon - 2\Lambda) - (3\sigma + 1)\kappa w_0 - N w_0 + \\ + \int a^*(x) \left[-\frac{\hbar^2}{2m^* \rho_0^2} \frac{d^2}{dx^2} \right] a(x) dx - G \int [a^*(x)a(x)]^2 dx. \quad (21)$$

With $G = 0$, to the extremum of the functional (21) correspond the values $a(x)$ with the form of plane waves

$$a(x) = \sqrt{\frac{1}{N}} \exp(ikx\rho_0). \quad (22)$$

With $G \neq 0$ we represent $a(x)$ in the form of modulated plane waves

$$a(x) = \varphi(x) \exp(ikx\rho_0), \quad (23)$$

where $\varphi(x)$ is a real function to be defined from the condition that formula (21) be a minimum, subject to the condition that the equality (18) be satisfied.

Substituting equation (23) into equation (21) we find

$$U(k, \varphi(x)) = -N w_0 + (\varepsilon - 2\Lambda) - (3\sigma + 1)\kappa w_0 + \\ + \frac{\hbar^2 k}{2m^* \rho_0^2} - \frac{\hbar^2}{2m^* \rho_0^2} \int \varphi(x) \frac{d^2 \varphi(x)}{dx^2} dx - G \int \varphi^4(x) dx. \quad (24)$$

Varying the functional (24) under the additional condition (18), we derive a non-linear equation

$$\frac{\hbar^2}{2m^* \rho_0^2} \frac{d^2 \varphi(x)}{dx^2} + 2G\varphi^3(x) - \chi\varphi(x) = 0 \quad (25)$$

to define $\varphi(x)$. Its solution that corresponds to the minimum of equation (24) at fixed $k \approx 0$, has the form

$$\varphi(x) = \sqrt{\mu/2} \operatorname{ch}^{-1}(\mu\xi), \quad (26)$$

where

$$\xi = x - x_0, \quad \mu \equiv \frac{2G}{\Delta E}, \quad \chi \equiv \frac{G^2}{\Delta E}, \quad (27)$$

and x_0 is the point on the chain where the maximum value of $\varphi(x)$ is concentrated. $\Delta E \approx 4\Lambda$ is the energy bandwidth of free excitons.

In the long wave approximation, we can put, in expression (13),

$$a_n a_{n+1}^* + a_n^* a_{n+1} \approx 2a_n^* a_n.$$

Going to a continuous model and making use of the values (26) we then obtain

$$\rho(\xi) \approx R_0 \left[1 - \frac{\sigma(\kappa + 2\eta)G}{\Delta E \cdot \text{ch}^2(\mu\xi)} \right]. \quad (28)$$

Consequently, a maximum shortening of the distance between peptide groups takes place at the point $x = x_0$. The distance decreases by the amount

$$\Delta\rho_{\max} = \sigma(\kappa + 2\eta)GR_0/\Delta E. \quad (29)$$

A substantial decrease in the distances between peptide groups should be observed for the length of the chain which contains $\Delta\xi$ peptide groups, where

$$\Delta\xi = \pi\Delta E/2G. \quad (30)$$

In this region the mean distance between peptide groups becomes

$$\rho_0 = R_0[1 - 0.6\sigma(\kappa + 2\eta)G/\Delta E]. \quad (31)$$

The total length of the chain in the excited state is determined by the expression

$$L \approx \left(1 - \frac{\sigma(\kappa + 2\eta)}{N} \right) \approx NR_0. \quad (32)$$

Substituting equation (26) into equation (24) we find the excitation energy (above the ground-state energy— Nw_0) for a fixed value k

$$E(k) = \left[\varepsilon - \frac{\Delta E}{2} - (3\sigma + 1)\kappa w_0 \right] + \frac{\hbar^2 k^2}{2m^*} - \frac{G^2}{3\Delta E}. \quad (33)$$

Thus, the excitation energy of a chain of resonantly interacting peptide groups of an α -helix is much less than the excitation energy ε of groups of isolated amino acid residues contained in the same cytoplasm.

Using equation (33) it is possible to calculate a group velocity of movement of the excitation along the α -helical molecule

$$V_g = \frac{1}{\hbar} \text{grad}_k E(k) = \frac{\hbar k}{m^*} = \frac{k}{\hbar} \Delta E \rho_0^2. \quad (34)$$

It follows from equation (34) that the velocity of movement of the contracted region along the molecule is proportional to the energy ΔE of the resonance interaction between peptide groups, and proportional to the quasi-momentum of the excitation. Parallel with the solution (26), equation (25) allows other solutions as well. However, at fixed k they do not correspond to the energy minimum of equation (24). For example, the solutions $\varphi = \sqrt{1/N}$ and $\chi = 2G/N$, which reduce equation (23) to plane waves, correspond to the excitation energy

$$E(k) = \left[\varepsilon - \frac{\Delta E}{2} - (3\sigma + 1)\kappa W_0 \right] + \frac{\hbar^2 k^2}{2m^*} - \frac{G}{N}.$$

Excitations of such a type are uniformly distributed along the whole length of the molecule.

We have considered the collective longitudinal excitations of one chain of peptide groups of an α -helical protein molecule. The excitation energy of this chain is less than 0.23 eV provided that it corresponds to the excitation of the valence vibration C=O. Therefore it appears likely that the disintegration energy of one ATP molecule is sufficient to excite one, two, and even all three chains formed by hydrogen bonds in the molecule.

To simplify calculations we have considered periodic boundary conditions, taking no account of excitation relaxation (attenuation) processes. When relaxation processes are taken into account, an excitation that arises at one end of a long molecule will spread to its other end at a decreasing rate, that is, it will give up a portion of its energy as it travels. At the same time the probability grows that the remaining portion of the excitation energy will all turn into heat before reaching the end of a molecule.

5. Excitation of Myosin Molecules and the "Sliding Model"

A quantum theory of the conformational change of a certain region of α -helical protein molecules under the excitation of their peptide groups has been developed in sections 3 and 4.† In the myofibrils of muscles such excitations may be generated by the disintegration energy of ATP molecules when, in the presence of Ca^{2+} , they interact with the active ATPase centres of the heads of the myosin molecules. The resulting collective excitation leads to shortening of the distances between peptide groups in some region of every hydrogen-bonded chain in the myosin molecule. As a result, in this region the helical myosin molecule thickens and bends, getting into a closer contact with the surrounding actin filaments.

The resulting collective excitation should travel from the head, where the ATPase centre is located, to the tail of the molecule. Since the frictional resistance (to motion) of the displaced twisted section of myosin filament relative to the surrounding actin filaments is considerably elevated, the movement of this "contracted" region from the head of the molecule to its tail will make the whole myosin filament move along actin filaments in the direction from the tail to the head (see Fig. 2). Such a "sliding" of a myosin filament over actin filaments will result in a slight shortening of the length of the I-discs of the sarcomere. The subsequent excitations that arise in the region

† This conformation corresponds in some respects to the possibilities discussed in the short paper by Volkenstein (1972). In that paper he introduced the notion "conformon" for excitation of the system of the long-wave phonons, electrons and local deformations of macromolecules.

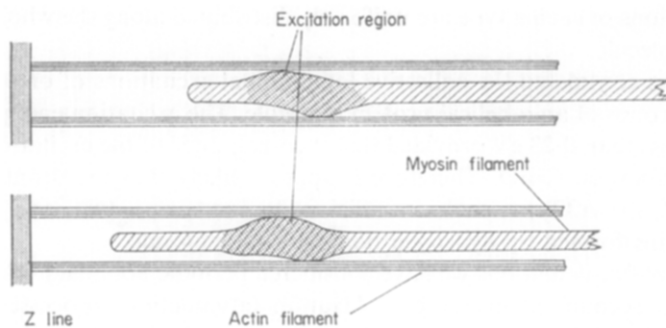


FIG. 2.

of the head (where the ATP is hydrolyzed in the presence of Ca^{2+}) and spread along myosin filaments to the sarcomere centre will lead to further movement of actin filaments with respect to myosin filaments. This movement resembles the movement of a snake ("concertina progression" (Gans, 1970)) in a straight narrow channel.

In the theory proposed here (and agreeing with the experimentally established "sliding model") the whole long helical protein molecule (and not only the head section as in Davies' theory) is the working element of the muscle. The second feature of the theory is that the "contracted" region spreads along the whole molecule, although an excitation initially arises in a separate part where the ATP energy is released.

So the main points of our theory are that: (1) the energy of hydrolysis of ATP is used in muscle to excite the carboxyl group at a peptide bond in the head of a myosin molecule; (2) this excitation is accompanied by a local distortion and contraction of the myosin molecule without gross overall changes in its length (see equation (32)); (3) the excitation with its attendant local contraction propagates down the molecule from head to tail because of resonance interaction between excited and non-excited peptide groups; and (4) this leads to a net movement of the myosin filament relative to the neighbouring actin filaments.

6. Some Numerical Estimates

The electric dipole moment of the vibration $\text{C}=\text{O}$ in a peptide group with frequency $\sim 1660 \text{ cm}^{-1}$, is equal to $d \approx 0.29$ Debye (Chirgadze & Rashevskaja, 1969). Therefore the radiation lifetime of the excited state with respect to spontaneous emission is

$$\tau = \frac{3\hbar}{4d^2} \left(\frac{c}{\omega} \right)^3 \approx 10^{-1} \text{ sec}; \quad \Delta E \approx 2 \times 10^{-15} \text{ erg.}$$

Excitons with kinetic energy equal $0.07 \Delta E$ have the wave number $k = 0.5/R_0$. They have the group velocity $V_g = \hbar k m^*/$, $m^* = 2\hbar^2/\Delta E \rho^2$, (Davydov, 1971, Ch. II, sec. 2), which is equal about 30 m/sec. When the deformability of a chain is taken into account, motion of the excitation is accompanied by a local distortion of the chain and this group velocity will decrease, as the effective mass m^* will increase. For quantitative description of this effect one must develop the theory in the next approximation which can take into account the motion of the peptide group. The velocity of the motion of the peptide group is not connected with the velocity of an exciton. The bigger the mass of a peptide group the smaller will be the role of deformation. If the mass of a peptide group is M_0 , the increase in the effective mass will be

$$m^* \Rightarrow m^*(1 - m^*B)^{-1},$$

where B is proportional $1/M_0$.

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